```
clc;
clear;
close all;
```

1. SIGNAL GENERATION

```
Fs = 8000;                  % Input sampling frequency (Hz)
f1 = 500;
f2 = 1500;                  % Signal components

M = 2;
L = 2;                      % Rate conversion factors
N = 1000;                   % Number of samples

fc_dec = 1800;
fc_interp = 3600;

n = 0:N-1;
t = n / Fs;

x = sin(2*pi*f1*t) + 0.8*sin(2*pi*f2*t);
```

2. DECIMATION AND INTERPOLATION

```
% Direct Downsampling
x_down_direct = x(1:M:end);
Fs_dec = Fs / M;

% Filtered Decimation
b_dec = fir1(60, fc_dec / (Fs/2));
x_dec_filtered = filter(b_dec, 1, x);
x_dec_filtered = x_dec_filtered(1:M:end);

% Upsampling and Filtering
x_up = zeros(1, length(x)*L);
x_up(1:L:end) = x;

Fs_interp = Fs * L;

b_interp = L * fir1(60, fc_interp / (Fs/2));
x_interp_filtered = filter(b_interp, 1, x_up);
```

3. COMMAND WINDOW DISPLAY

```
fprintf('================================================\n');
fprintf(' EXPERIMENT 12: MULTIRATE SIGNAL PROCESSING\n');
fprintf('================================================\n');

fprintf('Original Sampling Frequency (Fs): %d Hz\n', Fs);
fprintf('Decimation Factor (M): %d\n', M);
```

```
fprintf('New Sampling Frequency after Decimation: %d Hz\n\n', Fs_dec);

fprintf('Interpolation Factor (L): %d\n', L);
fprintf('New Sampling Frequency after Interpolation: %d Hz\n\n', Fs_interp);

fprintf('Signal Lengths:\n');
fprintf(' - Original Signal (x): %d samples\n', length(x));
fprintf(' - Decimated Signal (x_dec): %d samples\n', length(x_dec_filtered));
fprintf(' - Interpolated Signal (x_interp): %d samples\n\n',
length(x_interp_filtered));

fprintf('Status: Decimation and Interpolation executed successfully.\n');
fprintf('=================================================\n');
```

```
=================================================
 EXPERIMENT 12: MULTIRATE SIGNAL PROCESSING
=================================================
Original Sampling Frequency (Fs): 8000 Hz
Decimation Factor (M): 2
New Sampling Frequency after Decimation: 4000 Hz

Interpolation Factor (L): 2
New Sampling Frequency after Interpolation: 16000 Hz

Signal Lengths:
 - Original Signal (x): 1000 samples
 - Decimated Signal (x_dec): 500 samples
 - Interpolated Signal (x_interp): 2000 samples

Status: Decimation and Interpolation executed successfully.
=================================================
```

4. SPECTRAL COMPUTATION

```
Nfft = 2048;

[X_mag, f_x] = compute_spectrum(x, Fs, Nfft);

[X_down_mag, f_down] = compute_spectrum(x_down_direct, Fs_dec, Nfft);

[X_dec_mag, f_dec] = compute_spectrum(x_dec_filtered, Fs_dec, Nfft);

[X_up_mag, f_up] = compute_spectrum(x_up, Fs_interp, Nfft);

[X_interp_mag, f_interp] = compute_spectrum(x_interp_filtered, Fs_interp,
Nfft);

% Common Plot Parameters
figW = 8.5;
figH = 6.5;
```

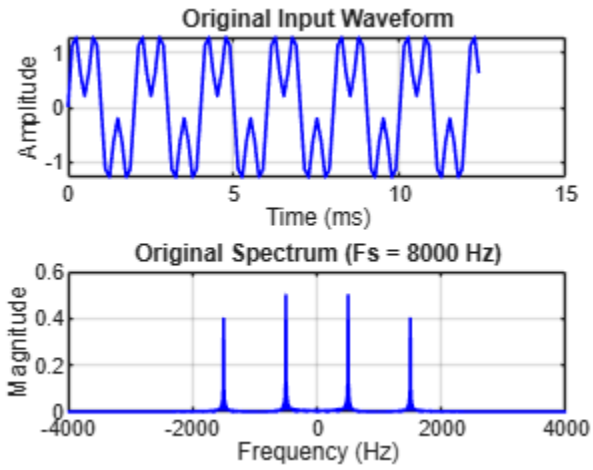
SUBFIGURE 12A: ORIGINAL SIGNAL

```
figA = figure('Units', 'centimeters', ...
    'Position', [2, 2, figW, figH]);

subplot(2,1,1);
plot(t(1:100)*1000, x(1:100), 'b', 'LineWidth', 1.2);
grid on;
title('Original Input Waveform');
xlabel('Time (ms)');
ylabel('Amplitude');

subplot(2,1,2);
plot(f_x, X_mag, 'b', 'LineWidth', 1.2);
grid on;
title('Original Spectrum (Fs = 8000 Hz)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');

exportgraphics(figA, 'Exp12_SubfigA_Original.png', 'Resolution', 600);
```



SUBFIGURE 12B: DECIMATED SIGNAL

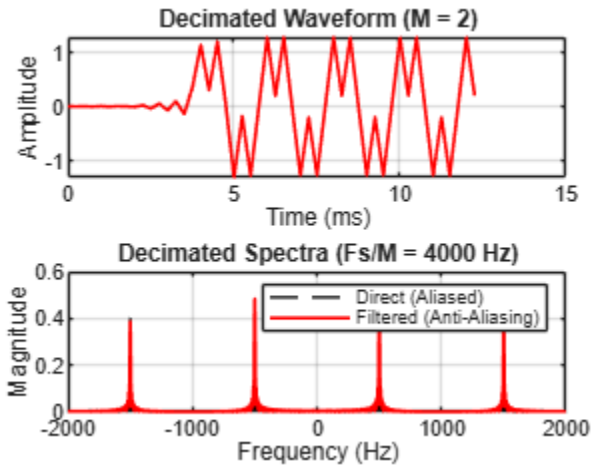
```
figB = figure('Units', 'centimeters', ...
    'Position', [11, 2, figW, figH]);

t_dec = (0:length(x_dec_filtered)-1) / Fs_dec;

subplot(2,1,1);
plot(t_dec(1:50)*1000, x_dec_filtered(1:50), ...
    'r', 'LineWidth', 1.2);
grid on;
title('Decimated Waveform (M = 2)');
xlabel('Time (ms)');
ylabel('Amplitude');
```

```
subplot(2,1,2);
plot(f_down, X_down_mag, 'k--', 'LineWidth', 1);
hold on;
plot(f_dec, X_dec_mag, 'r', 'LineWidth', 1.2);
grid on;
title('Decimated Spectra (Fs/M = 4000 Hz)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('Direct (Aliased)', 'Filtered (Anti-Aliasing)', ...
    'FontSize', 7);

exportgraphics(figB, 'Exp12_SubfigB_Decimated.png', 'Resolution', 600);
```



SUBFIGURE 12C: INTERPOLATED SIGNAL

```
figC = figure('Units', 'centimeters', ...
    'Position', [20, 2, figW, figH]);

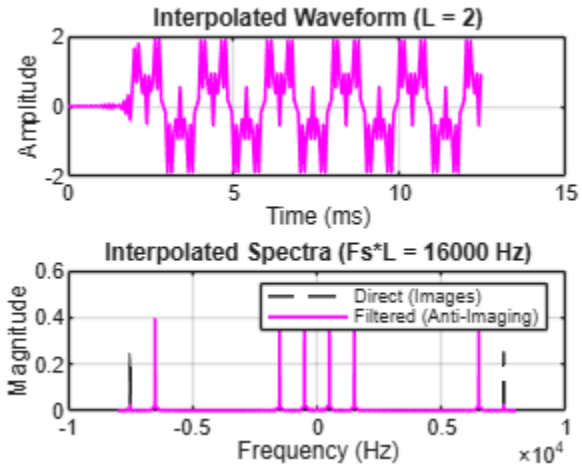
t_interp = (0:length(x_interp_filtered)-1) / Fs_interp;

subplot(2,1,1);
plot(t_interp(1:200)*1000, x_interp_filtered(1:200), ...
    'm', 'LineWidth', 1.2);
grid on;
title('Interpolated Waveform (L = 2)');
xlabel('Time (ms)');
ylabel('Amplitude');

subplot(2,1,2);
plot(f_up, X_up_mag, 'k--', 'LineWidth', 1);
hold on;
plot(f_interp, X_interp_mag, 'm', 'LineWidth', 1.2);
grid on;
title('Interpolated Spectra (Fs*L = 16000 Hz)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('Direct (Images)', 'Filtered (Anti-Imaging)', ...
```

```
    'FontSize', 7);

exportgraphics(figC, 'Exp12_SubfigC_Interpolated.png', 'Resolution', 600);
```



5. HELPER FUNCTION

```
function [X_mag, f] = compute_spectrum(signal, Fs_curr, Nfft)

    X = fft(signal, Nfft);
    X_mag = abs(fftshift(X)) / length(signal);

    f = (-Nfft/2 : Nfft/2 - 1) * (Fs_curr / Nfft);

end
```

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